Parth Kaushal

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Education

Vishwakarma Institute of Technology (VIT), Pune — Pune, India
Bachelor of Technology (B.Tech.) — Sep. 2025 – Jun. 2029
Computer Science and Engineering (IoT and CS, including BCT) — CGPA: 9.07/10.0

Experience

Drone Research & Development SIG (DRDS), VIT Pune, India
Avionics Subdivision Engineer (Mission 1 – RescueSwarm) — *Aug. 2026 – Present*

- Co-architecting the dual-drone RescueSwarm avionics stack (Scout and Delivery) for the upcoming NIDAR Mission 1 challenge, integrating C/C++ flight-control state machines on Pixhawk 6C with a Raspberry Pi 5 (13 TOPS AI HAT+) companion computer over MAVLink.
- Engineering multi-sensor fusion (NEO-F9P RTK GNSS, IMU, and rangefinder), RTL failsafes, 12S/60A ESC power distribution within a 19.68 kg AUW budget, dual isolated power rails, and servo-driven payload release with limit-switch feedback.

Technical Skills

Programming: C/C++, C#, Python, JavaScript, TypeScript
Web & Backend: Node.js, Express.js, FastAPI, PostgreSQL, Supabase, WebSockets, Vercel
Computer Vision & Data: OpenCV, PaddleOCR, face-api.js, scikit-learn, pandas
Development: ESP32, Git/GitHub, Electron, WebGL, Babylon.js, Unity, Unreal Engine, Phaser

Projects

Digital Metrology Inspector
Computer Vision & Automated Compliance Checks | SIH '26 Prototype

- Engineered a Python/OpenCV preprocessing pipeline using CLAHE and bilateral filtering to normalize label images under varying illumination before text extraction.
- Integrated PaddleOCR bounding-box detection with geometric font-dimension estimation and a regex-based rule engine to validate mandatory legal declarations against packaging requirements; advanced to the second internal selection round of SIH 2026.

Smart Helmet – RiderBeacon
IoT, Embedded Systems & Mobile Integration

- Developed ESP32 firmware to acquire six-axis MPU6050 motion data and stream telemetry to a Flutter mobile app over Bluetooth Low Energy (BLE) GATT characteristics.
- Implemented an asynchronous Flutter event flow for BLE parsing, phone-based GPS/location and speed tracking, and manual SOS signaling with emergency-contact settings and local state persistence.

FaceTrack – Offline-First Attendance System
Full-Stack Web Development & Edge Biometrics

- Architected a Svelte attendance application using face-api.js for in-browser face detection and 128-dimensional facial descriptor extraction, keeping biometric processing on the client.
- Implemented Dexie.js/IndexedDB persistence and a background synchronization path to transfer attendance records to a PostgreSQL/Express backend when connectivity returns.

ResonantAI – Neural Speech Restoration
Cross-Platform Desktop Application & Audio Processing | IIT Bombay Eureka!

- Developed an Electron desktop interface integrated with an asynchronous Python audio-processing service for speech-restoration workflows and audio-buffer handling.
- Organized model assets and Supabase-backed application workflows for session state and configuration; presented the project at IIT Bombay's Eureka! entrepreneurship competition.

Projects (Continued)

Nation-State
Browser-Based Simulation Game

- Engineered a TypeScript browser simulation with persistent player progression, game-state transitions, and dynamic economy and player-interaction logic.
- Integrated Supabase Google OAuth and session persistence; implemented achievement/badge progression and player report-card generation, and deployed the application on Vercel.

HardHit – 3D Cricket Game
WebGL & Interactive 3D Graphics

- Developed a TypeScript 3D cricket game with Babylon.js/WebGL, implementing scene hierarchy, camera controls, and interactive gameplay systems.
- Implemented bat-and-ball interaction, shot timing and trajectory logic, and physics-driven gameplay responses within a browser-rendered 3D environment.

IPL Predictor
Machine Learning & Data Engineering

- Built a pandas/scikit-learn pipeline to process historical ball-by-ball IPL data and prepare match-context features including teams, venue, and toss decision for outcome prediction.
- Integrated the prediction workflow with a Flask application to accept match inputs and return predicted outcomes.

Awards & Achievements

- 2026 – 1st Place, Stock Market Simulation, Chapters by IIT Bombay (Pune chapter).
- 2026 – 3rd Place, Case Competition, Chapters by IIT Bombay (Pune chapter).
- 2026 – Smart India Hackathon (SIH26): advanced to the second internal selection round with Digital Metrology Inspector.
- 2026 – Participant, IIT Bombay Eureka! with ResonantAI.

Technical Interests

Cybersecurity and privacy; AI/ML and computer vision; IoT and cyber-physical systems; decentralized systems and blockchain; full-stack and offline-first applications; game development, interactive 3D graphics, AR/VR, and spatial computing.